

Jianshen Liu

SOFTWARE ENGINEER · DISTRIBUTED STORAGE SYSTEM EXPERT

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Skills

DevOps	AWS/GCP, Jupyter Notebook, Docker, Kubernetes, Vagrant, Jenkins, Git, Mercurial, Maven, Gradle, VSCode, IntelliJ IDEA, CI/CD
Back-end	Linux Kernel, eBPF, DPDK, Ceph, Hadoop, MySQL, PostgreSQL, RocksDB, Apache Arrow, Ansible, REST API, NVMe, Microservices, Machine Learning, Distributed Systems
Programming Languages	C++, Python, Java, Bash, SQL, PHP, C#, Javascript, LaTeX, Vimscript
Others	English, Cantonese, Mandarin
Others	Patience, Open-Mindedness, Active Listening, Versatility, Enthusiasm, Agile Methodologies, Best Practices

Work Experience

Huawei Technologies Canada Co., Ltd.

Markham, Canada

STAFF ENGINEER

Apr 2025 – Present

- Boosted inference performance by 80% over Vanilla **vLLM** by optimizing the file-over-memory system and resolving critical E2E bottlenecks
- Delivered the file-over-memory project to the HQ platform team, managing the full lifecycle from research to final technology transfer
- Led code reviews and established quality standards, significantly accelerating performance verification and service stability

Huawei Technologies Canada Co., Ltd.

Markham, Canada

SENIOR ENGINEER

Aug 2023 – Mar 2025

- Engineered xPU data direct accelerator achieving 2.9 GB/s (73% utilization on PCIe Gen3), outperforming Nvidia's BaM solution by ~25%
- Slashed file-over-memory system latency by 90% by optimizing communication protocols between the client library and layout manager
- Developed the PD disaggregation feature for **vLLM**, enabling scalable AI acceleration and designing key session management modules for AI workload scheduler (AiWS)
- Led successful technical demos for major clients (e.g., Guangzhou Mobile), securing expanded research scope and investment from HQ
- Mentored 6+ engineers and co-ops on system design and complex I/O implementation, accelerating team delivery and code quality

University of California, Santa Cruz

Santa Cruz, USA

GRADUATE STUDENT RESEARCHER

Jan 2016 – Jul 2023

- Developing a **SmartNICs**-based **distributed data-flow management platform** for HPC big data workflows at Sandia National Laboratories
- Implemented *Bitar*, a **C++** library based on **Apache Arrow** and **DPDK**, to accelerate (>8x) data compression with hardware using the **BlueField-2 DPU**
- Characterized the potential of data processing using SmartNICs with microbenchmarks, **DPDK**, and an **in-kernel packet generator**; work highlighted by The Next Platform and Data Centre Dynamics
- Contributed features and bug fixes to multiple open-source projects, including **Ceph** and **Apache Arrow**
- Proposed a novel **media-based methodology** to quantify the benefits of offloading data access functions to storage devices, prototyped with **YCSB** and **RocksDB** on **ROCKPro64**, and automated with **Ansible** and **Python**
- Developed *no_fscache*, a **Linux kernel module** to address page cache pollution before RWF_UNCACHED was introduced into the POSIX API

KIOXIA America Inc.

San Jose, USA

STRATEGIC MARKETING ENGINEER INTERN

Jun 2020 - Sep 2020

- Quantified the performance speedup (>10x) of offloading SQL query processing to SSDs using **MariaDB**, **Docker**, **eBPF**, **FlameGraph**, **Linux perf**, **cgroups**, and the **TPC-H benchmark**
- Reported monthly to the senior vice president to drive internal strategic planning for **computational storage device** products
- Collaborated on cross-team projects to identify IO performance issues in NUMA systems

Samsung Semiconductor

San Jose, USA

PERFORMANCE ANALYSIS ENGINEER INTERN

Jul 2019 - Sep 2019

- Established a **distributed testing framework** to automate the performance evaluation for Samsung Network KV devices
- Created **eBPF**- and **kprobe**-based tools to expose the **performance observability** of the device API
- Identified major system bottlenecks with the tools developed, resolving issues boosted the write performance by 60%

Wumii Technology

Shenzhen, China

ANDROID DEVELOPMENT TEAM LEAD

Dec 2012 - Jun 2014

- Led design and code reviews of the first **reading recommendation app** (among the top 25 social apps) in China
- Implemented an energy-efficient and highly available Paho **MQTT**-based notification module in **Java** that outperformed commercial solutions in terms of latency by 35%

SENIOR SOFTWARE ENGINEER

Jun 2010 - Dec 2012

- Worked as a core engineer to develop the first **news recommendation system** in China
- Drove the system architecture to handle distributed requests based on the **Spring** and **Hibernate** framework, and the code obfuscation system using Google's **Closure Compiler**
- Responsible for maintaining high availability for all backend services
- Launched a **Java**-, **PHP**-, and **Javascript**-based platform to present picturized article recommendations across multiple platforms, achieved 200 million monthly active users

Education

University of California, Santa Cruz

Santa Cruz, USA

DOCTOR OF PHILOSOPHY (PH.D.), COMPUTER SCIENCE AND ENGINEERING

2016 - Present

PhD Candidate advised by Prof. Carlos Maltzahn with a GPA of 3.96/4.00

Dissertation topic: **Extending Composable Data Services to the Realm of Embedded Systems**

South China Agricultural University

Guangzhou, China

BACHELOR OF SCIENCE (BS), INFORMATION AND COMPUTING SCIENCE

2006 - 2010

GPA: 3.45/4.00

Honors & Awards

May 2023 **Best Paper Award**, 2nd Workshop on Composable Systems (COMPSYS '23)

Florida USA

Sep 2022 **Outstanding Student Paper**, 26th Annual 2022 IEEE High Performance Extreme Computing (HPEC '22)

Virtual Conference

Oct 2009 **National Encouragement Scholarship**, South China Agricultural University

Guangzhou, China

Oct 2009 **The Second Prize Scholarship**, South China Agricultural University (top 10%)

Guangzhou, China

Oct 2008 **State Grants**, South China Agricultural University

Guangzhou, China

Oct 2008 **The Second Prize Scholarship**, South China Agricultural University (top 10%)

Guangzhou, China

Publications

Extending Composable Data Services into SmartNICs

Craig Ulmer, Jianshen Liu, Carlos Maltzahn, Matthew L. Curry

2nd Workshop on Composable Systems (COMPSYS '23), Co-located with IPDPS 2023, 2023, Florida USA

Processing Particle Data Flows with SmartNICs

Jianshen Liu, Carlos Maltzahn, Matthew Leon Curry, Craig Ulmer

26th Annual 2022 IEEE High Performance Extreme Computing (IEEE-HPEC 2022), 2022, Virtual Conference

Performance Characteristics of the BlueField-2 SmartNIC

Jianshen Liu, Carlos Maltzahn, Craig Ulmer, Matthew Leon Curry

Sandia National Lab. (SNL-CA), Livermore, CA (United States); Sandia National Lab. (SNL-NM), Albuquerque, NM (United States) (May 2021). 2021

Implementing a Kernel Module to Eliminating External Caching Effects

Jianshen Liu

The Center for Research in Open Source Software (CROSS) (Jan. 2020). 2020

Scale-out Edge Storage Systems with Embedded Storage Nodes to Get Better Availability and Cost-Efficiency At the Same Time

Jianshen Liu, Matthew Leon Curry, Carlos Maltzahn, Philip Kufeldt

MBWU: Benefit Quantification for Data Access Function Offloading

Jianshen Liu, Philip Kufeldt, Carlos Maltzahn

International Conference on High Performance Computing (HPC-IODC 2019), 2019, Springer, Cham

Presentations & Posters

Experience of using the BlueField-2 DPU

Presented at Sandia National Laboratories (May 2022). 2022

Processing Particle Data Flows with SmartNICs

Presented at UC Santa Cruz Open Source Symposium (Sept. 2022). 2022

Eusocial Storage Devices

Presented at KIOXIA America Inc., Seagate Technology, and Fujitsu America Inc. (2020). 2020

MBWU (MibeeWu): Quantifying benefits of offloading data management to storage devices

Poster Session at 17th USENIX Conference on File and Storage Technologies (FAST'19) (Feb. 2019). 2019

Quantifying benefits of offloading data management to storage devices

Poster Session at 2019 OCP Global Summit Symposium (Mar. 2019). 2019

Certificates

Jan 2022 **CruzHacks Mentor Certificate**, CruzHacks

Santa Cruz, USA

Dec 2020 **Graduate Student Professional Communication Certificate**, University of California, Santa Cruz

Santa Cruz, USA

Program Committees

CruzHacks

Santa Cruz, USA

TECH MENTOR

Jan 2020, Jan 2021

Mentored hackers to build web applications to solve real-world problems

Center for Research in Open Source Software (CROSS)

Santa Cruz, USA

WORKSHOP CHAIR

Oct 2021, Oct 2020

Chaired the SmartNICs and Eusocial Storage Devices workshop at the 5th and 6th CROSS Research Symposium at UC Santa Cruz